USED MEDIAS LLC
EMBEDDED SYSTEMS DIVISION

Boot Chain and Image Format

UM-NATOS-002

DOCUMENT	REVISION	STATUS	CLASSIFICATION
UM-NATOS-002	1.0 · 2026-08-14	current — measured on hardware	Internal

1. Abstract

This report documents the complete path from power-on to the first instruction of nat-os, the binary format the kernel image must satisfy, and the measured boot trace from the target board. It answers the question "how does this C code actually run on the device" precisely enough to debug a failure to boot.

2. The four stages

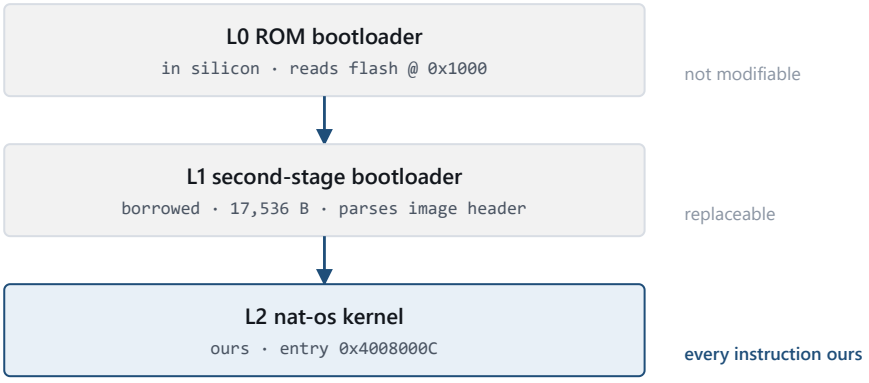
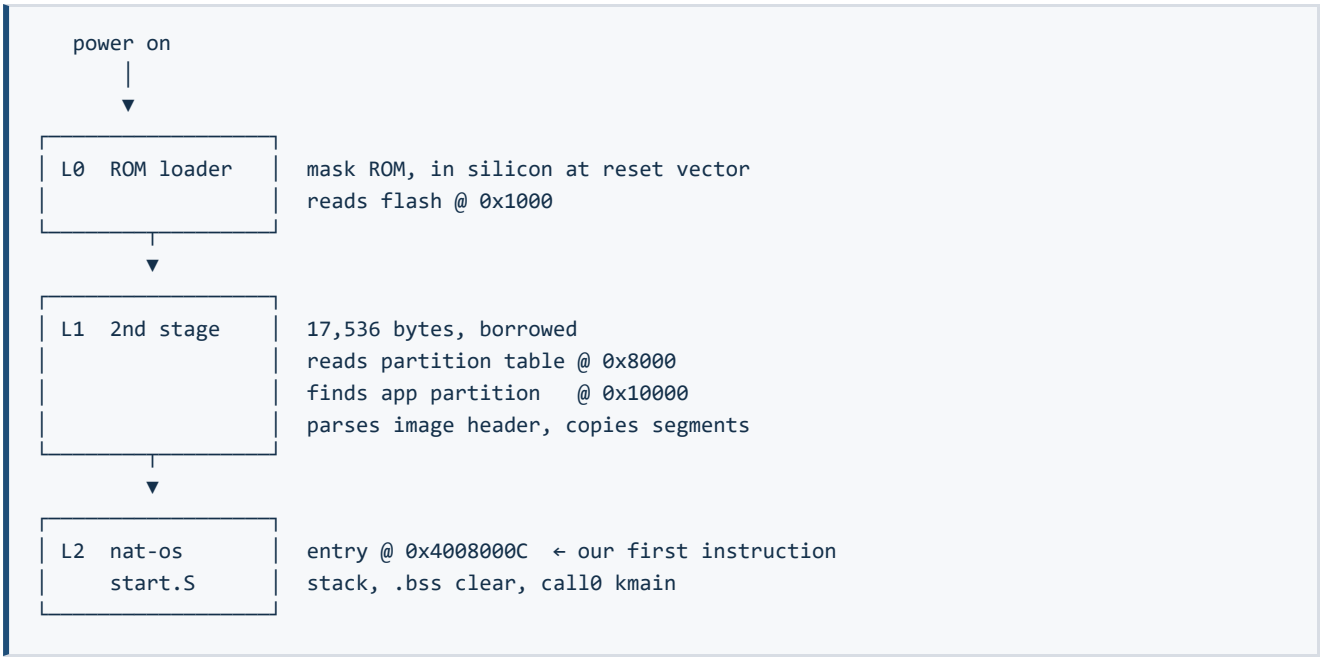


Figure 1. *The four boot stages. Only L2 upward is project code; the interface to L1 is the image header alone.*



2.1 Stage L0 — mask ROM

Executes from ROM on reset. Determines boot mode from strapping pins (GPIO0 high = normal flash boot), configures the SPI flash interface, and loads the second-stage bootloader from flash offset `0x1000`. It also brings up UART0 at 115200 baud for its own diagnostic output — a side effect nat-os currently depends on (see §6).

Not modifiable.

2.2 Stage L1 — second-stage bootloader

Borrowed from the CYD PlatformIO project. Responsibilities relevant to us:

1. Read the partition table at `0x8000`.
2. Locate the application partition (offset `0x10000`).
3. Read the image header at that offset.
4. Copy each declared segment to its declared load address.
5. Verify the image checksum and SHA-256 hash.
6. Jump to the declared entry point.

The interface between L1 and nat-os is entirely the image header. Nothing else is shared — no symbols, no runtime, no calling convention. This is why the bootloader is replaceable without touching kernel code.

2.3 Stage L2 — nat-os

Begins at `_start` in `kernel/start.S`. See UM-NATOS-003 for why this code can assume a call0 environment.

3. Image format

`esptool elf2image` converts the linked ELF into the format L1 expects. The produced header declares, for each segment, a length and a load address, plus one entry point for the whole image.

Measured output for the current build:

```
Image version:  1
Entry point:    0x4008000C
2 segments
  Segment 1: len 0x00188  load 0x3FFB0000  file_offs 0x018  [BYTE_ACCESSIBLE, DRAM]
  Segment 2: len 0x002E0  load 0x40080000  file_offs 0x1A8  [IRAM]
Checksum:         0x8F (valid)
Validation hash:  8e3a272a…5c72f3 (valid)
```

Both load addresses correspond directly to the `MEMORY` regions declared in `kernel/linker.ld`. The entry point is `_start + 0x0C` rather than `+0x00` because the assembler places the literal pool at the head of the section; see UM-NATOS-005 §4.

There is no magic in this format. It is a list of "copy N bytes to address A", followed by "jump to address E". Any producer that emits a conforming header will boot.

4. Measured boot trace

Captured 2026-08-14 from the target board on COM5 at 115200 baud, with the serial port opened *before* reset so no output was missed.

```
ets Jul 29 2019 12:21:46

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1184
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4

====================================
 nat-os  milestone 0 — kernel alive
====================================
```

4.1 Reading the trace

Line	Meaning
`ets Jul 29 2019`	ROM build date — identifies the silicon revision's ROM
`rst:0x1 (POWERON_RESET)`	Clean cold boot. **Diagnostically important:** a watchdog or panic reset shows a different code here, and is the first thing to check when a kernel change stops booting
`boot:0x13 (SPI_FAST_FLASH_BOOT)`	Normal flash boot; GPIO0 was high
`mode:DIO, clock div:2`	Flash interface configuration
`load:0x…` ×3	**The bootloader loading itself**, not nat-os
`entry 0x400805e4`	Entry into the *bootloader*, not our kernel

The three `load:` lines and the `entry` line are frequently misread as the kernel loading. They are not. They are L1 placing its own segments before it has even read our partition. Our own load is silent — L1 does not log it — and the first evidence of nat-os is the banner.

4.2 Verified handoff

The banner appearing at all proves the header was well-formed, the checksum matched, both segments were copied, and the CPU reached `0x4008000C`. The self-check lines that follow prove more; see UM-NATOS-006.

5. Flash layout

Offset	Contents	Size	Origin
`0x1000`	Second-stage bootloader	17,536 B	Borrowed
`0x8000`	Partition table	3,072 B	Borrowed
`0x10000`	nat-os kernel image	1,216 B	**Ours**

Everything from `0x10000` onward is available to the project. The two borrowed regions total under 21 KB.

6. Known dependencies on L1 behaviour

These are places where nat-os currently relies on the bootloader or ROM having done something, and would break if L1 were replaced naively.

1. **UART0 is already configured.** The kernel writes bytes into the TX FIFO without setting baud rate, pin mux, or line discipline. This works because the ROM configured UART0 at 115200 for its own output. A replacement L1 that did not do this would produce a silent kernel. *Resolution: implement UART configuration in the kernel — small, and removes the dependency.*
2. **Flash cache is configured but unused.** M0 executes entirely from RAM. When the kernel outgrows IRAM it will begin depending on the cache mapping L1 set up, which is currently unexamined.
3. **CPU clock.** Running at whatever L1 selected. No kernel code reads or sets the clock, so all current timing (including the M0 delay loop) is of unknown absolute rate.

7. Failure modes and first diagnostic step

Symptom	Most likely cause
No output at all	Image checksum invalid, or entry point outside a loaded segment
ROM trace then silence	Kernel reached but faulted before UART output — suspect stack or `.bss`
Boot loop with `rst:0x3` / `rst:0xc`	Watchdog — kernel hung before feeding or disabling it
Garbled output	Baud mismatch — the ROM's rate was changed or the monitor disagrees

In every case the **reset reason on the first line** is the highest-value single datum, which is why the capture procedure resets the board explicitly rather than attaching to an already-running system.

8. Reproduction

```
cd C:\Users\nobod\Projects\nat-os
.\build.ps1 -Flash -Port COM5
# then capture with the port opened before reset
```

See UM-NATOS-005 for the capture harness and why attaching after reset loses the banner.

9. References

- UM-NATOS-004 — Memory Map (segment addresses and the overlap risk)
- UM-NATOS-005 — Build and Flash Pipeline
- UM-NATOS-006 — Milestone 0 Verification Report